OPERATIONAL THREAT INTELLIGENCE REPORT

Underwater Image Enhancement and Threat Detection Analysis

Report ID:	TEST_001_2025
Operation Timestamp:	27 November 2025, 14:30:00 IST
Mission Coordinates:	19.0760°N, 72.8777°E
Deployment Depth:	45 meters
Classification:	RESTRICTED

ISSUING AUTHORITY: Defence Research and Development Organisation (DRDO)
DISTRIBUTION: Authorized Personnel Only
VALIDITY: Operational Intelligence - Time Sensitive

SECTION A: OPERATIONAL METADATA

Mission Parameters

Timestamp: 27 November 2025, 14:30:00 IST
GPS Coordinates: 19.0760°N, 72.8777°E
Camera Deployment Depth: 45 meters
Water Turbidity Score: 6.8/10
Visibility Conditions: Moderate
Water Temperature: 24.5°C
Current Speed: 1.8 knots

SECTION B: THREAT INTELLIGENCE ANALYSIS

ALERT: Total 2 potential threat(s) detected and analyzed. Detailed assessment follows:

THREAT #1

Object Type:	submarine
Confidence Level:	92.0%
Distance from Camera:	150.5 meters
Risk Assessment Score:	8/10
Priority Level:	HIGH
Detection Method:	YOLOv8-X Deep Learning
Bounding Box (x,y,w,h):	[100, 200, 300, 400]

THREAT #2

Object Type:	boat
Confidence Level:	87.0%
Distance from Camera:	85.2 meters
Risk Assessment Score:	6/10
Priority Level:	MEDIUM

Detection Method:	YOLOv8-X Deep Learning
Bounding Box (x,y,w,h):	[400, 150, 600, 350]

SECTION C: VISUAL INTELLIGENCE REPORT

C.1 Image Enhancement Analysis

Advanced deep learning-based image enhancement was applied to improve underwater visibility and threat detection accuracy. The enhancement process corrects color attenuation, removes scattering effects, and restores lost visual information.

C.2 Threat Detection Overlay

C.3 AI Explainability - Grad-CAM Analysis

Gradient-weighted Class Activation Mapping (Grad-CAM) visualizes the attention regions used by the AI model for threat detection. Multi-scale analysis incorporates edge detection, superpixel segmentation, visual saliency, color distinctiveness, and texture richness to provide comprehensive model interpretability.

C.4 Comprehensive Enhancement Analysis Grid

The 12-panel analysis grid provides detailed examination of enhancement performance across multiple dimensions: intensity correction, RGB channel restoration, color balance, contrast enhancement, brightness adjustment, texture preservation, edge definition, and quantitative underwater image quality metrics.

C.5 Image Quality Metrics

Metric	Value	Assessment
PSNR (Peak Signal-to-Noise Ratio)	28.5 dB	Good Quality
SSIM (Structural Similarity Index)	0.89	Good Similarity
UIQM (Underwater Image Quality Measure)	2.85	Good UW Quality
Turbidity Reduction	65.3%	Improvement
Entropy Gain	0.245	Information Recovery

SECTION D: MISSION SUMMARY AND RECOMMENDATIONS

Mission Summary

Underwater surveillance operation conducted at depth 45 meters. AI-enhanced image processing detected 2 potential threats in the operational area.

Recommended Operator Actions

- Maintain continuous surveillance of detected threat locations
- Verify threat classification with secondary sensors
- Report findings to naval operations command center

Severity Rating

HIGH

REPORT STATUS: COMPLETE
VERIFICATION: AI-Generated Intelligence Report
AUTHORIZATION: Aqua Sentinel Automated Threat Detection System